FMCG Supply Chain Optimisation

Predictive Modelling for Optimal Warehouse Shipment Quantity

Final Project Report

Great Learning — PGP Data Science & Business Analytics

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1. Understanding the Business Problem

1.1 Problem Statement

A fast-moving consumer goods (FMCG) company that entered the instant noodles market two years ago faces significant supply chain imbalances across its national warehouse network. In certain regions, product demand far exceeds available supply, causing stockouts and lost sales; in others, excess inventory accumulates and deteriorates, causing product wastage and inflated holding costs. This mismatch is both operationally disruptive and financially costly in a highly competitive market characterised by thin margins.

The data science objective is to build a predictive regression model using historical operational and sales data to determine the optimal quantity of product (in tons) to ship to each warehouse per quarter. The target variable is product_wg_ton — the total weight of product shipped to a warehouse over the last three months.

1.2 Need for the Study

Without a data-driven mechanism, supply allocation relies on heuristics or simple averages that cannot account for the diverse operational, locational, and infrastructural characteristics of individual warehouses. The consequences are twofold: overstock elevates storage costs and product expiry risk; understock in high-demand regions drives customers to competitors, permanently eroding market share. A predictive model trained on warehouse-level operational data can forecast optimal quantities with far greater precision.

1.3 Business and Social Opportunity

The business opportunity is threefold. First, accurate demand forecasting reduces inventory losses by aligning supply with actual regional demand, improving working capital efficiency. Second, improved allocation eliminates missed sales in high-demand zones. Third, regional demand patterns identified by the model can inform targeted advertising campaigns in high-potential catchment areas — directly driving market share growth in the instant noodles segment. Socially, reduced food wastage from more precise supply planning has meaningful environmental and sustainability benefits.

2. Exploratory Data Analysis

2.1 Methodology

All EDA was conducted in Python using Pandas for data manipulation, Matplotlib and Seaborn for visualisation, and NumPy for numerical operations, within a Google Colab environment. The analysis followed a structured three-stage approach: univariate analysis examined each feature's individual distribution via count plots, histograms, and boxplots; bivariate analysis explored pairwise relationships between features and the target variable using scatter plots, grouped bar charts, and box plots; and multivariate analysis assessed feature interactions jointly through a correlation heatmap and pairplot. All figures were generated programmatically in the submitted notebook and are referenced sequentially as Figures 1–17 in this report. Note that plot titles within the notebook may use different internal labels; figures should be cross-referenced by notebook cell number rather than plot title.

2.2 Data Collection and Background

The dataset was sourced from the company's internal warehouse management system and contains historical records for 25,000 warehouses covering a three-month operational snapshot. Each row represents one warehouse. The dataset includes 24 columns capturing operational, logistical, infrastructural, geographical, and regulatory attributes. The target variable product_wg_ton — total product weight shipped in tons over the last three months — contains no missing values, ensuring there is no target-side data quality issue.

2.3 Data Overview

2.3.1 Dataset Shape and Structure

The dataset contains 25,000 rows and 24 columns. Of these, 8 are categorical (object dtype) and 16 are numerical (14 int64, 2 float64). Two identifier columns (Ware_house_ID, WH_Manager_ID) each have 25,000 unique values with no predictive signal and were dropped prior to modelling.

Table 1: *Dataset Summary Statistics*

Attribute	Value
Total records	25,000
Total features	24 (including target)
Numerical features	16 (14 int64, 2 float64)
Categorical features	8 (object)
Target variable	product_wg_ton — continuous, range 2,065–55,151 tons
Mean / Std target	22,103 tons / 11,608 tons
Duplicate rows	0 — confirmed clean
Missing values	3 columns: wh_est_year (47.5%), workers_num (4.0%), approved_wh_govt_certificate (3.6%)

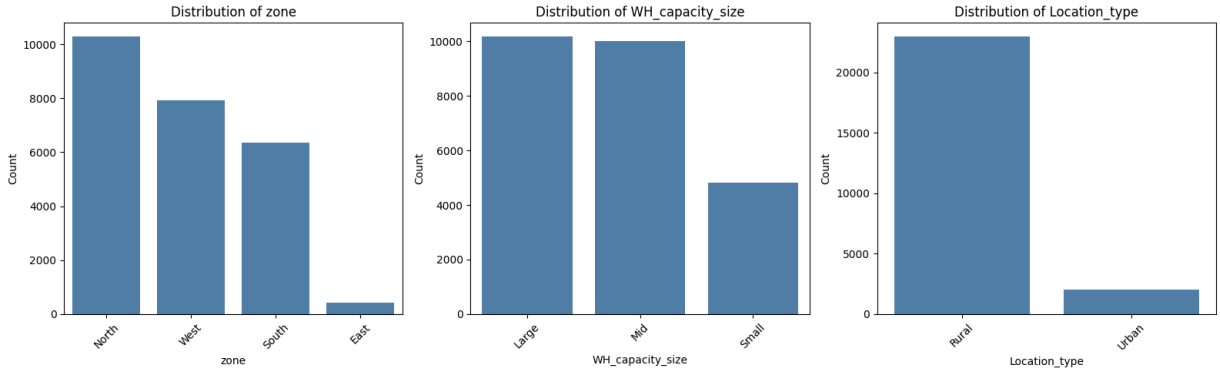
2.3.2 Descriptive Statistics of Numerical Features

Table 2: *Descriptive Statistics — Key Numerical Features*

Feature	Min	Q1	Median	Q3	Max	Mean	Std
product_wg_ton (TARGET)	2,065	13,059	22,101	30,103	55,151	22,103	11,608
dist_from_hub (km)	55	109	164	218	271	163.5	62.7
workers_num	10	24	28	33	98	28.9	7.9
retail_shop_num	1,821	4,313	4,859	5,500	11,008	4,986	1,053
num_refill_req_l3m	0	2	4	6	8	4.1	2.6
transport_issue_l1y	0	0	0	1	5	0.77	1.2
wh_breakdown_l3m	0	2	3	5	6	3.5	1.7
govt_check_l3m	1	11	21	26	32	18.8	8.6
Competitor_in_mkt	0	2	3	4	12	3.1	1.1
distributor_num	15	29	42	56	70	42.4	16.1

2.4 Univariate Analysis

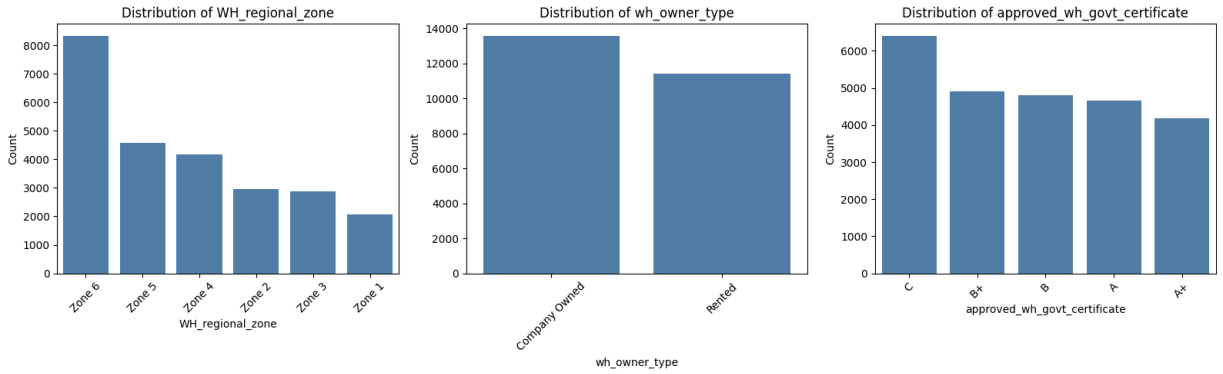
2.4.1 Categorical Feature Distributions



Count plots for zone, WH_capacity_size, and Location_type

Notebook source: Notebook Cell 15

Figure 1: *Distribution of Zone, Warehouse Capacity Size, and Location Type*



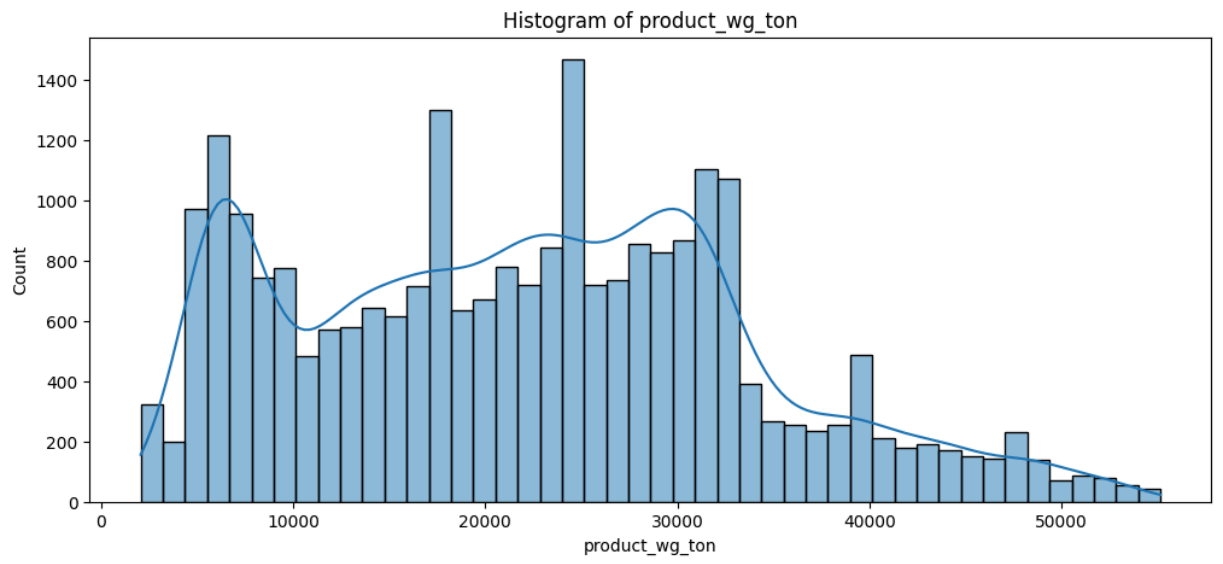
Count plots for WH_regional_zone, wh_owner_type, and approved_wh_govt_certificate

Notebook source: Notebook Cell 16

Figure 2: *Distribution of Regional Zone, Owner Type, and Government Certificate Grade*

Figures 1 and 2 reveal several notable patterns. Location_type is heavily skewed towards Rural (91.8%), indicating the company's supply chain is predominantly structured around rural distribution infrastructure. WH_capacity_size shows a near-equal split between Large (~10,200) and Mid-sized (~10,100) warehouses with fewer Small ones (~4,700), suggesting a preference for higher-capacity storage. Zone distribution reveals a strong concentration in North (~10,300) and severe underrepresentation of East (~500 warehouses) — a geographic imbalance that likely contributes to regional demand-supply mismatches. Zone 6 dominates the regional zone distribution (~8,300 warehouses). wh_owner_type is nearly balanced with Company Owned at 54.3% vs. Rented at 45.7%. Grade C (lowest quality tier) is the most prevalent certification grade (~6,700 warehouses), raising quality concerns across the network.

2.4.2 Target Variable Distribution



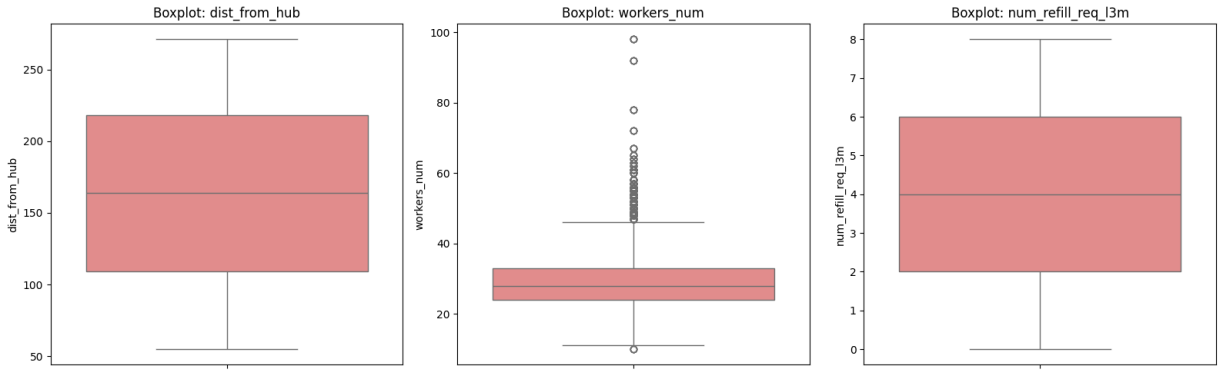
Histogram with KDE overlay for product_wg_ton

Notebook source: Notebook Cell 17

Figure 3: *Distribution of Product Shipment Quantity (product_wg_ton) — Target Variable*

The target variable product_wg_ton exhibits a multimodal distribution with a mild right skew, ranging from 2,065 to 55,151 tons with a mean of 22,103 tons. Multiple distinct peaks are visible at approximately 5,000–8,000, 18,000, 25,000, and 30,000–32,000 tons. This multi-peak pattern suggests warehouses naturally cluster into distinct shipment tiers, most likely determined by capacity category. The right skew reflects a small number of very high-volume warehouses pulling the mean above the median. This variability confirms that a single uniform shipment quantity is wholly inappropriate — precise, warehouse-level prediction is essential for this problem.

2.4.3 Numerical Feature Distributions — Outlier Detection

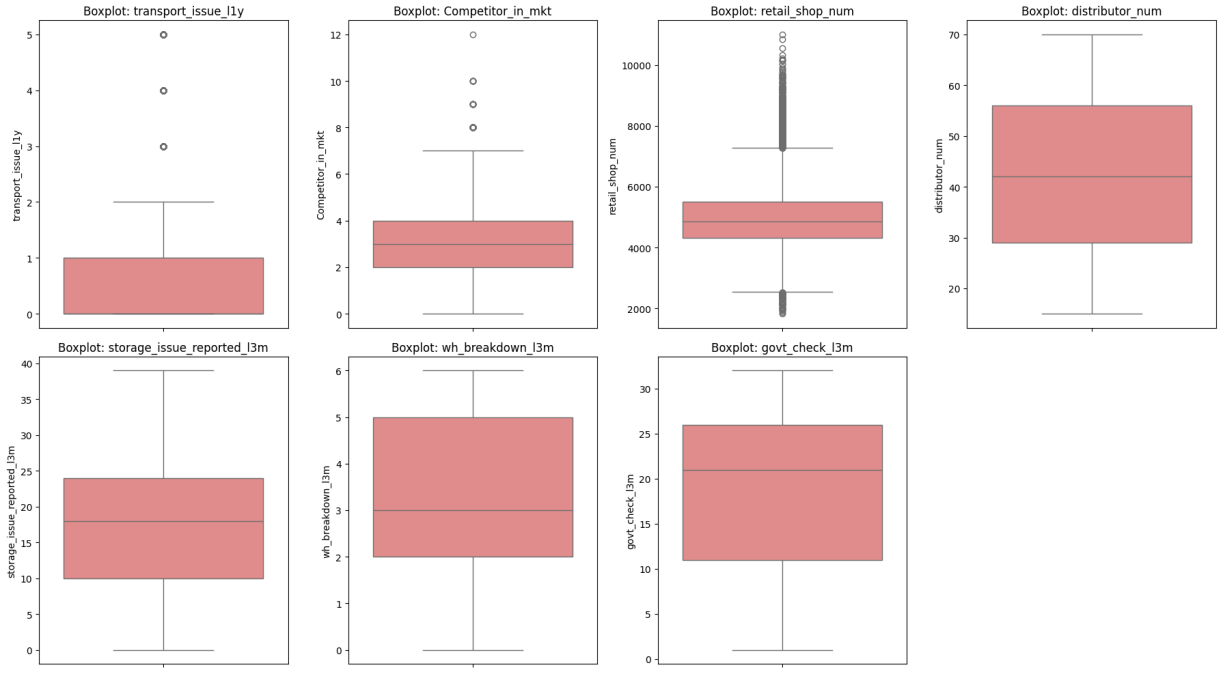


Boxplots of dist_from_hub, workers_num, and num_refill_req_l3m

Notebook source: Notebook Cell 18

Figure 4: *Boxplots of dist_from_hub, workers_num, and num_refill_req_l3m*

dist_from_hub: Clean, symmetric distribution (55–271 km, median 164 km) with no outliers, confirming consistent geographic spread across the network. num_refill_req_l3m: Also clean (0–8, median 4), suggesting stable restocking demand. workers_num: Tight IQR of 24–33 workers but with multiple high outliers extending to 98, indicating a small number of unusually large warehouses; these were subsequently treated via IQR capping.



Boxplots of transport_issue_l1y, Competitor_in_mkt, retail_shop_num, distributor_num, wh_breakdown_l3m, govt_check_l3m

Notebook source: Notebook Cell 19

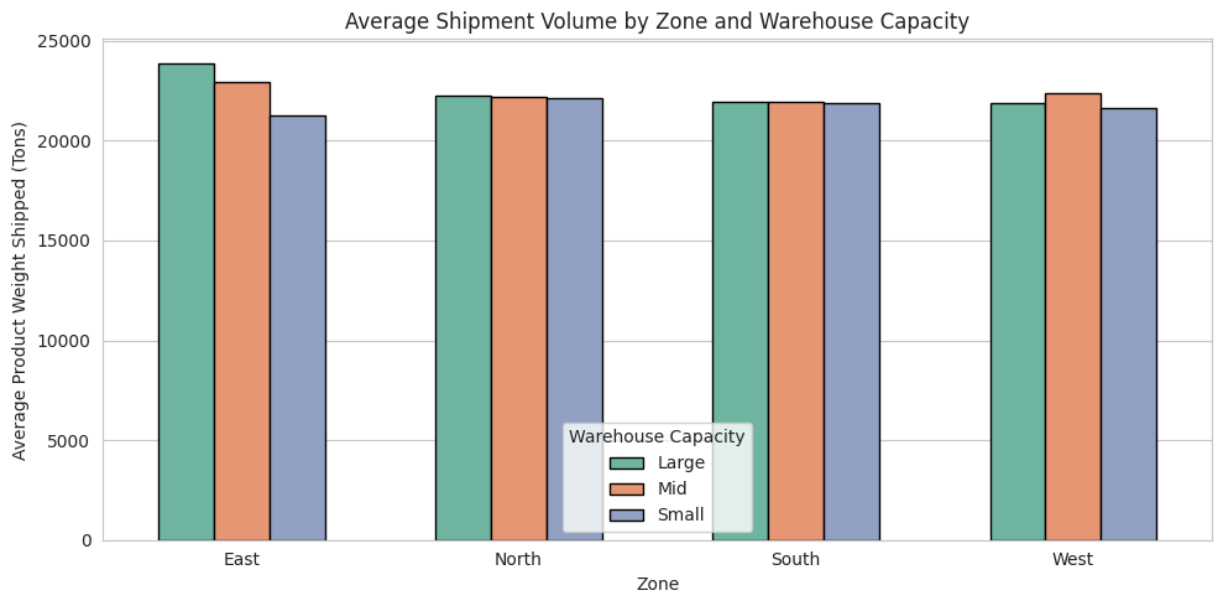
Figure 5: *Boxplots of Remaining Numerical Features (full in Appendix B)*

transport_issue_l1y: Outliers at values 3–5; the IQR of 0–1 means warehouses reporting 3+ incidents per year are statistically extreme. Competitor_in_mkt: Outliers above 7; IQR is 2–4. retail_shop_num: Severe outliers on both ends (below ~2,000 and above ~7,500), reflecting extreme variation in the retail networks served. distributor_num, wh_breakdown_l3m, and govt_check_l3m exhibit clean distributions without significant outliers.

2.5 Bivariate Analysis

The bivariate analysis examines the relationship between each feature and the target variable product_wg_ton to identify potential predictors and guide feature selection.

2.5.1 Shipment Volume by Zone and Warehouse Capacity



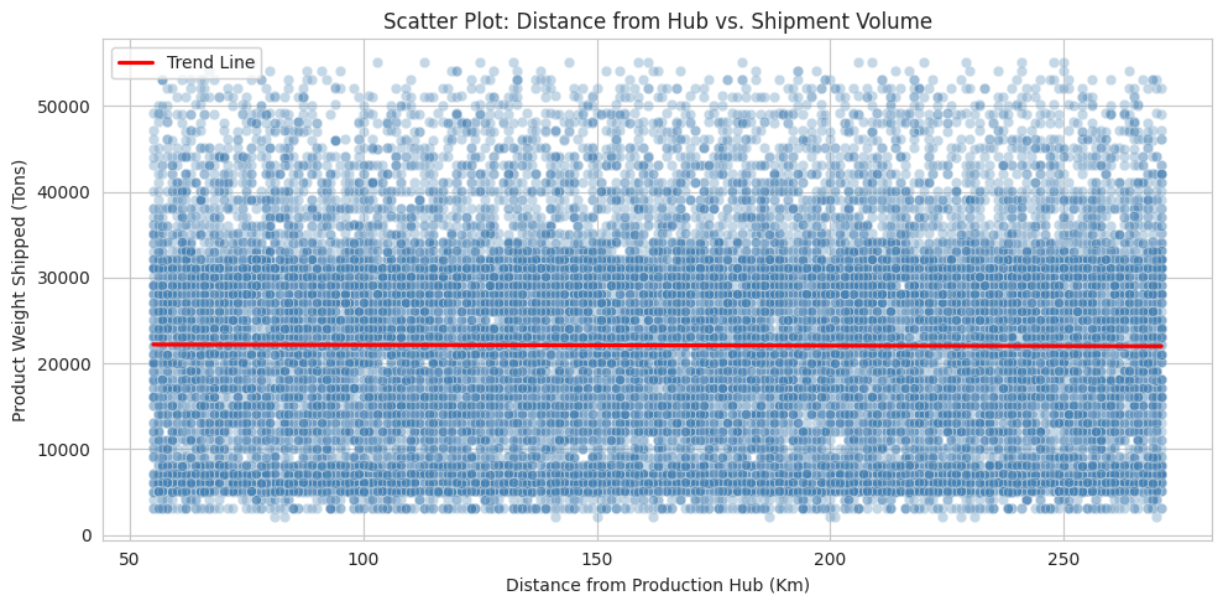
Grouped bar chart: Mean product_wg_ton by zone and WH_capacity_size (hue)

Notebook source: Notebook Cell 20

Figure 6: *Average Shipment Volume by Zone and Warehouse Capacity Size*

Average shipment volumes are broadly consistent across all four zones (approximately 21,000–24,000 tons), indicating that neither geographic zone nor warehouse capacity is a strong independent predictor of shipment quantity. The East zone shows the most pronounced size effect for Large warehouses (~24,000 tons), while the West zone exhibits an anomaly where Mid-sized warehouses slightly outperform Large ones — potentially reflecting local demand dynamics. The overall uniformity across zones suggests the company's current allocation model does not effectively differentiate by region.

2.5.2 Distance from Hub vs. Shipment Volume



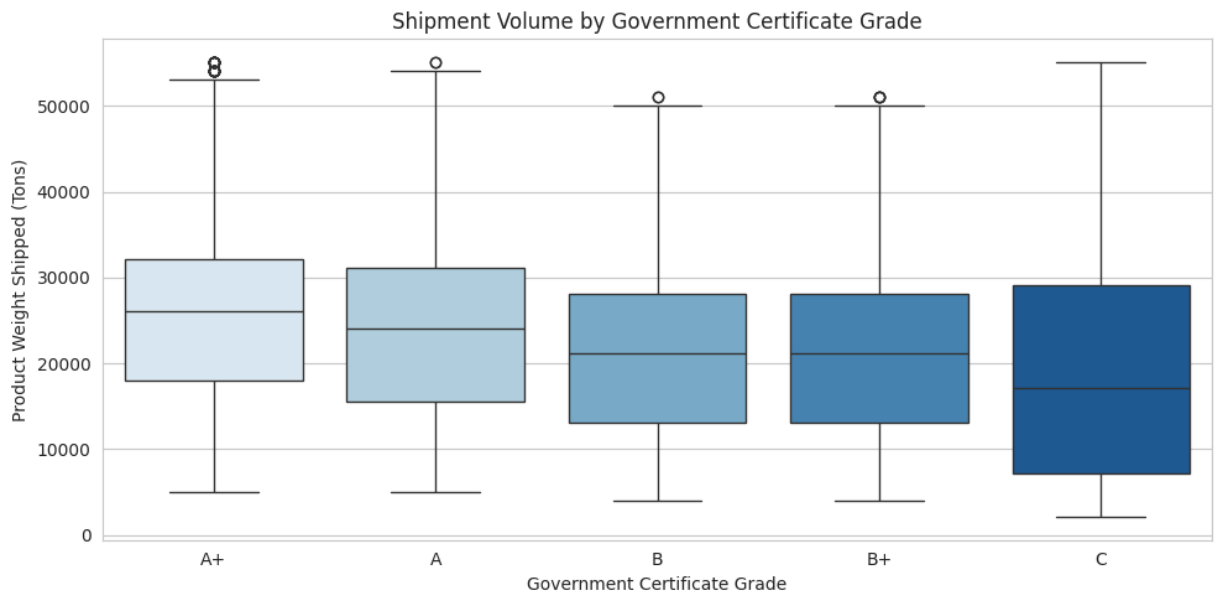
Scatter plot with OLS regression trend line: dist_from_hub vs. product_wg_ton

Notebook source: Notebook Cell 21

Figure 7: *Distance from Production Hub vs. Shipment Volume*

The near-horizontal regression trend line confirms no meaningful linear relationship between warehouse distance from the production hub and shipment volume. Shipments are uniformly distributed across the full distance range (55–271 km), regardless of proximity to the production facility. Ideally, logistics cost optimisation should result in higher allocations to closer warehouses; the absence of this pattern represents a cost-efficiency opportunity.

2.5.3 Shipment Volume by Government Certificate Grade



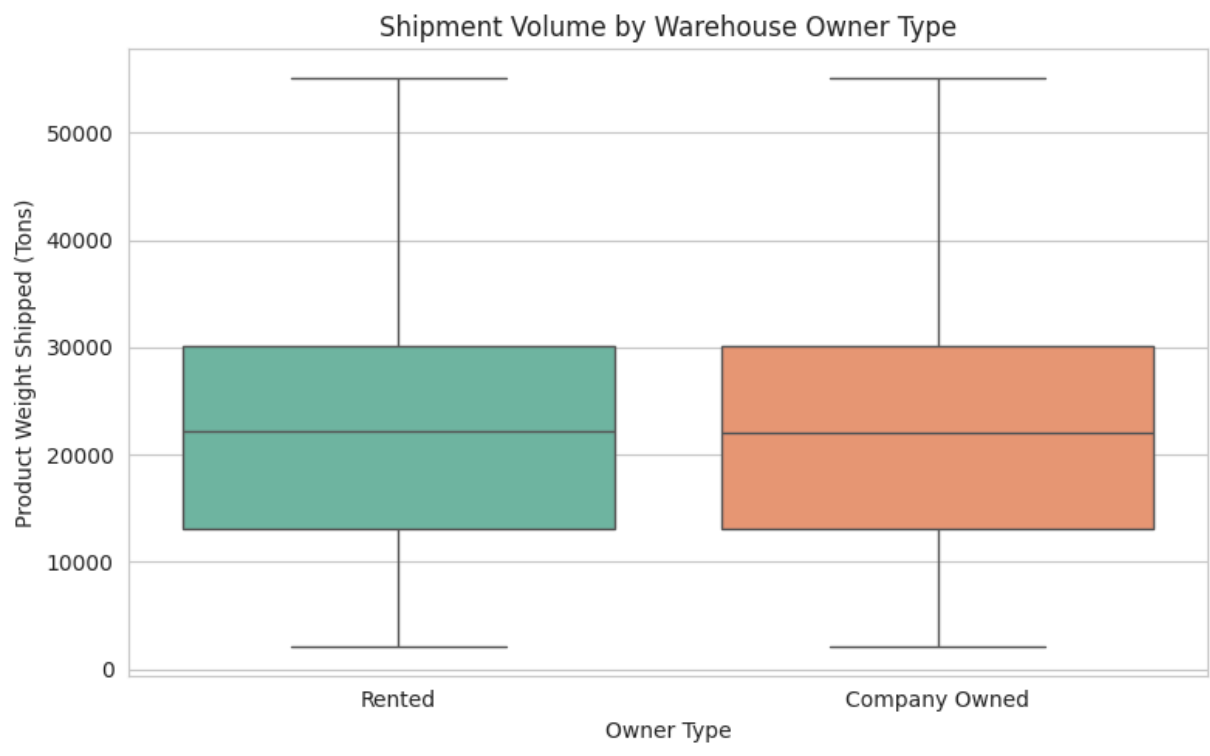
Box plots: product_wg_ton by approved_wh_govt_certificate grade (ordered A+ to C)

Notebook source: Notebook Cell 22

Figure 8: *Shipment Volume by Government Certificate Grade*

A clear positive monotonic relationship exists between certification grade and shipment volume. A+ and A grade warehouses achieve median shipments of approximately 25,000–26,000 tons, while Grade C warehouses (the most common) show a median of approximately 17,000 tons — a difference of ~9,000 tons per quarter. This confirms the company allocates more product to better-certified, higher-quality facilities, which is operationally sound but creates a quality-investment incentive.

2.5.4 Shipment Volume by Warehouse Owner Type



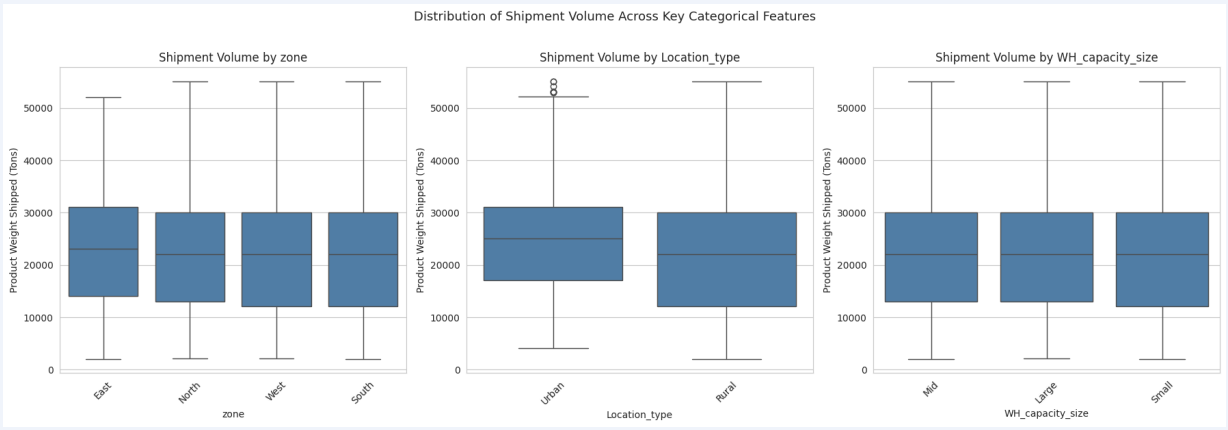
Box plots: product_wg_ton by wh_owner_type (Company Owned vs. Rented)

Notebook source: Notebook Cell 23

Figure 9: *Shipment Volume by Warehouse Owner Type*

Both Company Owned and Rented warehouses show nearly identical median shipment volumes (~22,000 tons) and similar interquartile ranges. Ownership type exerts negligible influence on supply allocation decisions — confirming that operational performance characteristics, not ownership structure, are what matter.

2.5.5 Shipment Volume by Location Type, Zone, and Capacity



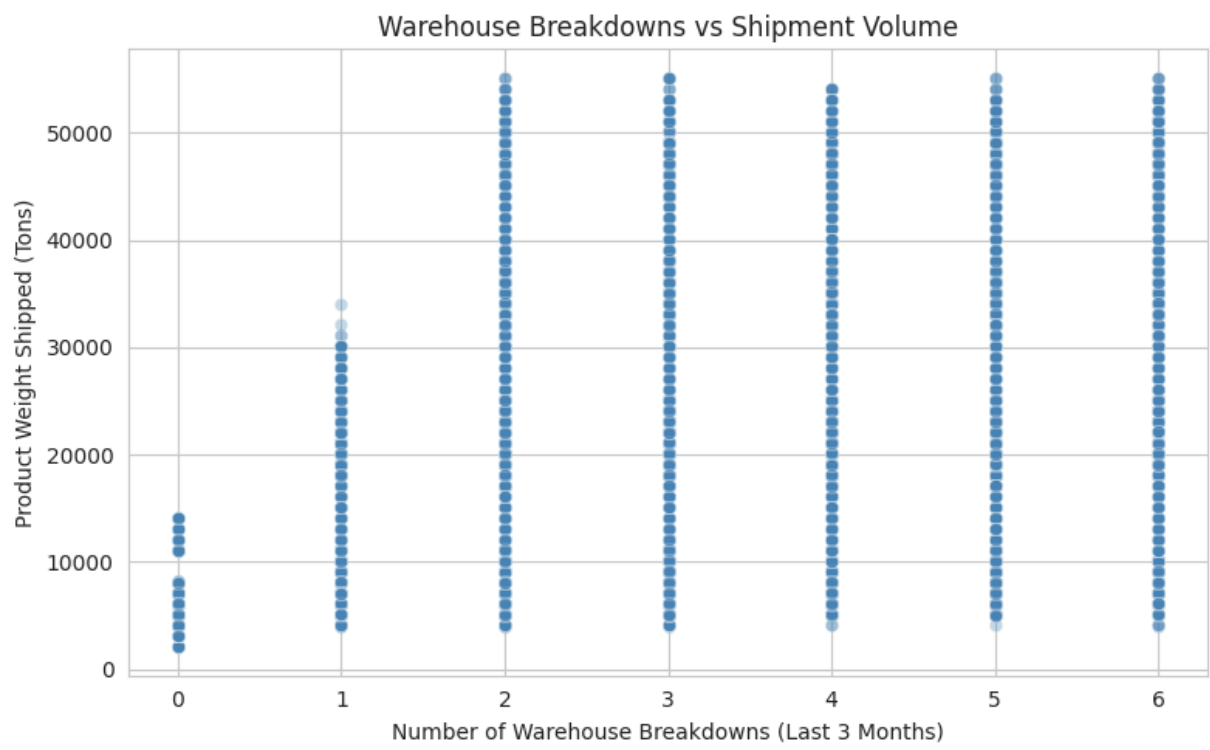
Side-by-side box plots: product_wg_ton by zone, Location_type, and WH_capacity_size

Notebook source: Notebook Cell 24

Figure 10: *Shipment Volume Across Key Categorical Features*

All four geographic zones show near-identical median shipments (~22,000–23,000 tons) with heavily overlapping IQRs, further confirming zone is not an independent predictor. Large, Mid, and Small warehouses also show similar distributions, indicating warehouse capacity size alone does not govern allocation. Urban warehouses show a slightly higher median (~25,000 tons) and a compressed lower tail compared to Rural, suggesting urban locations receive marginally higher and more consistent allocations.

2.5.6 Warehouse Breakdowns vs. Shipment Volume



Scatter plot: wh_breakdown_l3m (x-axis) vs. product_wg_ton (y-axis)

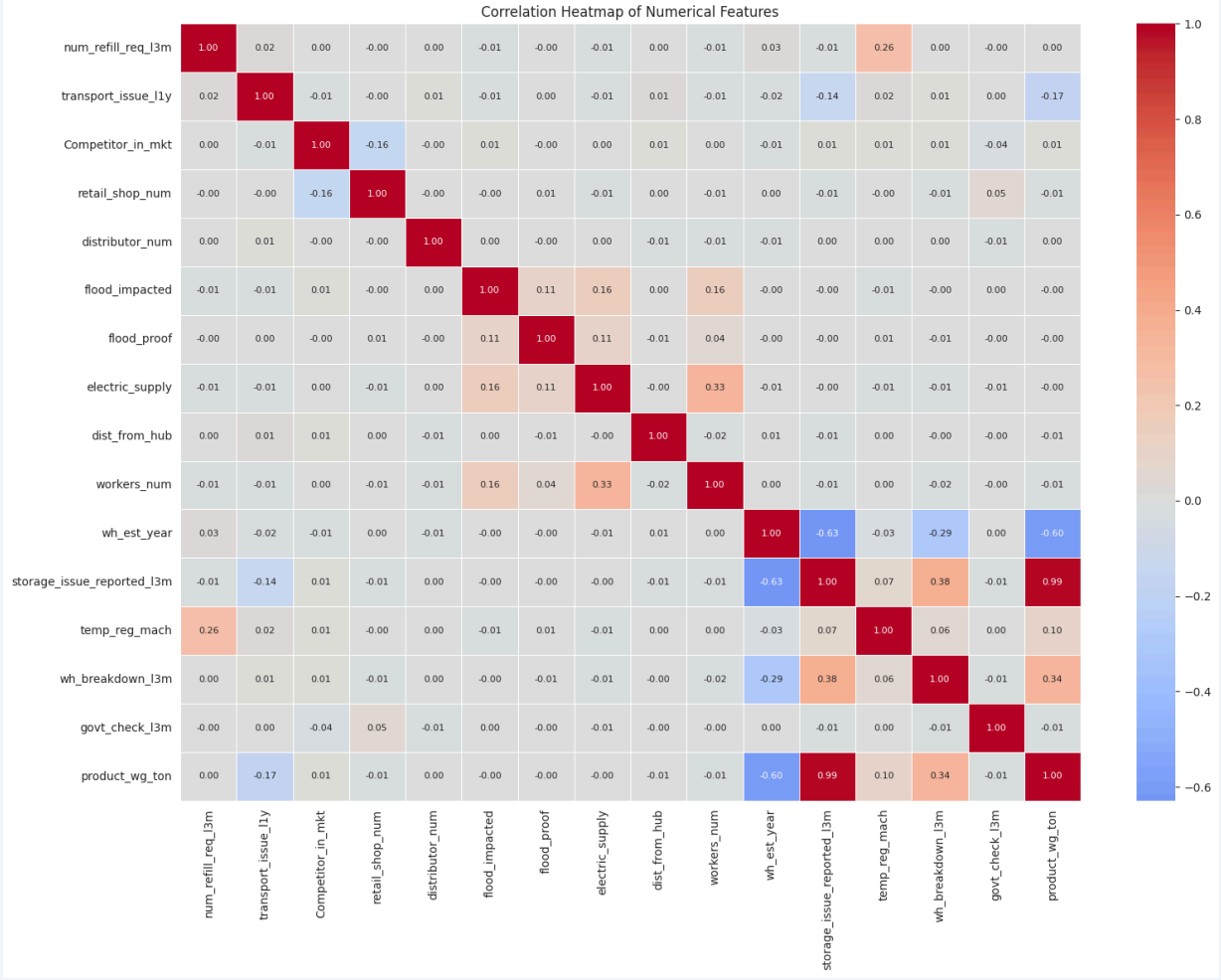
Notebook source: Notebook Cell 25

Figure 11: *Warehouse Breakdowns (Last 3 Months) vs. Shipment Volume*

This scatter plot reveals the most striking bivariate pattern in the entire dataset. Warehouses reporting zero breakdowns are confined to a narrow low-volume band (roughly 2,000–15,000 tons), while those with two or more breakdowns span the full range up to 55,000 tons. This strongly suggests breakdown frequency is a reliable proxy for operational intensity — high-throughput, high-volume warehouses experience more operational strain. This feature later emerged as the dominant predictor in the XGBoost model, accounting for ~52% of total importance.

2.6 Multivariate Analysis

2.6.1 Correlation Heatmap



Annotated correlation heatmap of all numerical features (coolwarm palette, annot=True)

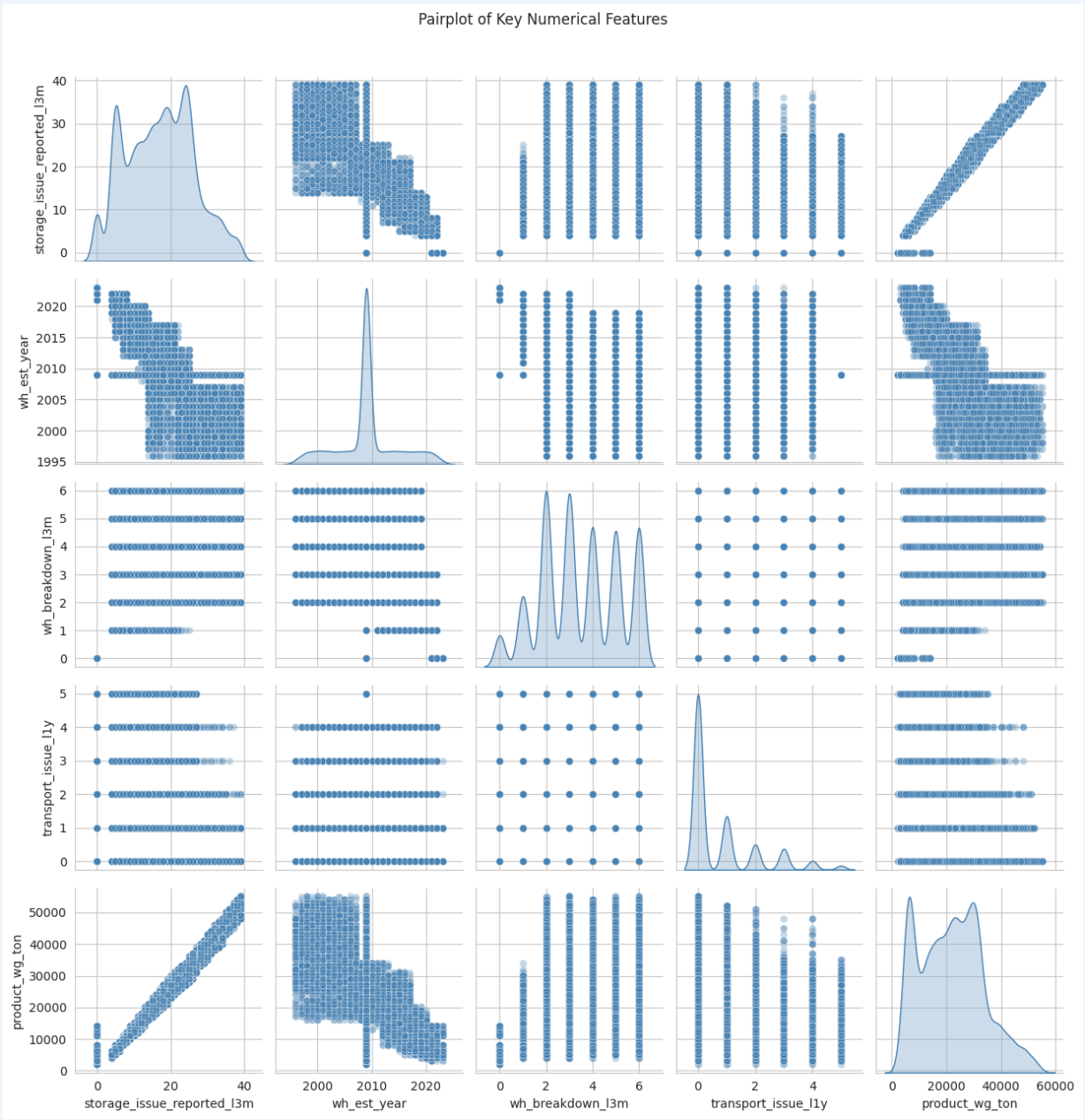
Notebook source: Notebook Cell 26

Figure 12: *Correlation Heatmap of Numerical Features*

The heatmap reveals two critical findings that directly shaped the preprocessing strategy. First, storage_issue_reported_l3m has a near-perfect Pearson correlation of r = 0.99 with product_wg_ton — an immediate data leakage flag confirmed in preprocessing. Second, wh_est_year shows a moderate negative correlation of r = −0.60 with the target; older, more established warehouses consistently handle larger shipment volumes, likely reflecting greater operational experience and established distribution networks. wh_breakdown_l3m shows a moderate positive correlation of r = 0.34, consistent with the bivariate finding. The majority of features show very weak linear correlations with the target (|r| < 0.15), suggesting that non-linear relationships and interactions — which tree-based models can capture — will be important for achieving good predictive performance.

Among inter-feature correlations: wh_est_year and storage_issue_reported_l3m show r = −0.63 (older warehouses report more storage issues — structurally logical given ageing infrastructure). No other feature pair shows concerning multicollinearity (all |r| < 0.5 among remaining predictors).

2.6.2 Pairplot of Key Features



Pairplot: storage_issue_reported_l3m, wh_est_year, wh_breakdown_l3m, transport_issue_l1y, product_wg_ton

Notebook source: Notebook Cell 27

Figure 13: *Pairplot of Key Numerical Features (full version in Appendix B)*

The pairplot provides visual confirmation of correlation heatmap findings. The near-perfect linear diagonal between storage_issue_reported_l3m and product_wg_ton is the most striking feature, visually confirming data leakage. The descending staircase between wh_est_year and product_wg_ton confirms older warehouses ship more. wh_breakdown_l3m shows clear banding by shipment level — its relationship is non-linear and step-like, which linear models cannot fully exploit. transport_issue_l1y shows no discernible pattern against any feature, consistent with its weak but statistically significant negative coefficient in OLS.

2.7 Key Insights and Observations from EDA

The following eight insights emerge from the exploratory analysis. Each directly informs either the modelling strategy or the business recommendations in Sections 4–8.

- Insight E1 — Data Leakage Confirmed: storage_issue_reported_l3m has r = 0.99 with the target. Its ratio with product_wg_ton is nearly constant across all 25,000 records (range: 1,009–1,345×), confirming it is a mathematically derived quantity, not an independent predictor. It must be dropped unconditionally before modelling.
- Insight E2 — Warehouse Operational Experience Is a Strong Predictor: wh_est_year (r = −0.60 with target) indicates older warehouses handle significantly more product — likely due to established distribution relationships, proven logistics networks, and accumulated operational efficiency.
- Insight E3 — Breakdown Frequency Signals Throughput Intensity: wh_breakdown_l3m (r = 0.34) exhibits a clear step-change with shipment volume. Warehouses with zero breakdowns are exclusively low-volume; high-volume warehouses invariably report multiple breakdowns. This feature is the dominant predictor in the final model (52% XGBoost importance).
- Insight E4 — Transport Reliability Directly Reduces Supply Allocation: Higher transport_issue_l1y values consistently correlate with lower shipment quantities. The OLS model quantifies this at −2,170 tons per incident — the most controllable negative driver in the supply chain.
- Insight E5 — Government Certification Grade Drives Allocation Confidence: The ~9,000 ton per quarter gap between Grade A+ and Grade C median allocations confirms the company assigns more product to higher-quality, better-certified facilities. A four-step upgrade (C→A+) is associated with approximately +7,500 tons per quarter per warehouse.
- Insight E6 — Zone and Capacity Have Minimal Independent Predictive Power: Neither geographic zone nor warehouse capacity size independently differentiates shipment allocations. The current supply model appears not to reward larger warehouses or specific regions with systematically higher allocations.
- Insight E7 — Target Variable is Multimodal — Distinct Demand Tiers Exist: The multiple peaks in the product_wg_ton distribution confirm warehouses naturally cluster into distinct shipment tiers. A single average-based allocation strategy is inappropriate for all warehouses; the model must learn these tier boundaries.
- Insight E8 — Distance from Hub Is an Untapped Cost Lever: The complete absence of correlation between dist_from_hub and shipment volume suggests logistics cost is not currently factored into allocation. This is an actionable efficiency opportunity — distance-weighted allocation adjustments could reduce logistics spend without compromising service levels.

3. Data Preprocessing

3.1 Methodology

The final preprocessing pipeline (Cells F1–F4) is built around a single principle: the train-test split is performed before any statistical fitting. All imputation parameters (medians, modes), outlier fences, and scaling parameters are computed exclusively on the 20,000-row training set and then applied to both train and test sets. This prevents contamination of the model's evaluation environment and ensures all test set performance metrics reflect genuine generalisation to unseen data.

An earlier interim pipeline (Cells 1–40) did not follow this discipline — it performed preprocessing on the full 25,000-row dataset before splitting — introducing subtle data leakage and producing inflated performance figures. The final pipeline corrects this, and the interim pipeline's results are retained in the notebook for reference but are not used for evaluation.

3.2 Duplicate Value Check and Treatment

A row-level duplicate check (df.duplicated().sum()) returned 0 duplicate records across all 25,000 rows. Data integrity is confirmed at the record level — no treatment was required.

3.3 Anomalous Value Check and Treatment

An anomalous value check was performed on all numerical columns. wh_est_year was the primary candidate for invalid entries (future establishment years would be anomalous). The column ranges from 1996 to 2023 — all values are historically valid. No other numerical column contained values outside plausible operational ranges. No treatment was required.

3.4 Missing Value Check and Treatment

Three columns contain missing values. Following the train-test split, imputation parameters were fitted on the training set (20,000 rows) only and applied to both partitions:

Table 3: *Missing Value Treatment Summary*

Column	Missing Count	Missing %	Treatment Applied	Rationale
workers_num	990 total (808 in train)	~4.0%	Median imputation (train median = 28)	Outliers present; median more robust than mean
approved_wh_govt_certificate	908 total (725 in train)	~3.6%	Mode imputation (train mode = 'C')	Categorical; most frequent category is the safest fill
wh_est_year	11,881 total	~47.5%	Dropped (final pipeline)	Median imputation assigns ~12,000 records the same value (2009), severely biasing the feature

The decision to drop wh_est_year from the final pipeline was taken after observing that median imputation at 47.5% missingness assigned the same year (2009) to approximately 12,000 rows — making warehouse_age (derived from this column) appear artificially predictive in the interim pipeline. Dropping it is the most statistically honest treatment given the severity of missingness.

3.5 Outlier Check and Treatment

Outlier detection was performed visually using boxplots (whis=1.5) and confirmed via the IQR method. Four columns were found to contain outliers. Treatment applied: IQR Capping (Winsorization) — values below

Q1−1.5×IQR or above Q3+1.5×IQR are clipped to the respective fence values. This preserves all 25,000 rows while removing the distorting influence of extreme values. Fence parameters were computed on the training set and applied identically to the test set.

Table 4: *Outlier Capping Summary*

Column	Lower Fence	Upper Fence	Outliers Treated	Business Rationale
workers_num	10.50	46.50	607	Most warehouses employ 24–33 workers; caps unusual operations
transport_issue_l1y	−1.50	2.50	2,943	Most warehouses have 0–1 incidents/year; 3+ are extreme
Competitor_in_mkt	−1.00	7.00	96	IQR 2–4; a few warehouses face unusually high competition
retail_shop_num	2,543	7,271	948	Extreme variation in retail density; typical IQR 4,313–5,500

3.6 Feature Engineering

One derived feature was created in the interim pipeline: warehouse_age = 2026 − wh_est_year, converting the establishment year into an operationally meaningful metric representing warehouse operational experience (range: 3–30 years, mean: 16.80 years). This transformation is conceptually sound and performed well in the interim pipeline. However, since wh_est_year was dropped in the final pipeline due to severe missingness, warehouse_age was not created in the final pipeline. The feature is retained in Cells 1–40 of the notebook for reference.

3.7 Data Leakage Handling

storage_issue_reported_l3m was identified as a data leakage feature and dropped from the very start of the final pipeline (Cell F1). Three lines of evidence confirm the leakage:

- Pearson correlation with target: r = 0.987 — the highest correlation of any feature, far exceeding all genuine predictors.
- Near-constant ratio: product_wg_ton / storage_issue_reported_l3m remains within an extremely narrow band of 1,009–1,345 across all 25,000 records. No genuinely independent predictor could maintain such consistency with the target.
- Structural inference: Storage issues being reported at corporate are almost certainly generated from shipment-related operational data — the column is likely a derived reporting metric, not an independent measurement. Including it would allow any model to trivially reconstruct the target, producing inflated metrics that would not generalise.

3.8 Class Imbalance Handling

Class imbalance handling is not applicable to this project. product_wg_ton is a continuous numerical target variable, making this a regression problem. Class imbalance is a concern in classification tasks with disproportionate class frequencies; it does not apply here. All 25,000 target values are present and valid. No treatment was required.

3.9 Categorical Encoding

Six categorical features were encoded based on their measurement scale:

- Ordinal encoding: WH_capacity_size → {Small:0, Mid:1, Large:2} and approved_wh_govt_certificate → {C:1, B:2, B+:3, A:4, A+:5}. Both columns have a natural rank ordering that should be preserved numerically.
- Binary encoding: Location_type → {Urban:1, Rural:0} and wh_owner_type → {Company Owned:1, Rented:0}.
- One-hot encoding with drop_first=True: zone (4 categories → 3 dummies) and WH_regional_zone (6 categories → 5 dummies). These are purely nominal; drop_first eliminates the dummy variable trap by dropping one reference category per feature.

After encoding, the feature matrix expanded from 18 columns to 25 columns.

3.10 Feature Scaling

StandardScaler was applied, transforming each feature to mean ≈ 0 and standard deviation ≈ 1. The scaler was fitted exclusively on the 20,000-row training set and then applied to both training and test sets — ensuring no leakage of test set statistics into the normalisation parameters. Scaling is primarily required for the OLS linear regression baseline (where feature scale directly affects coefficient magnitudes and gradient descent convergence), and is less critical for tree-based models which use the unscaled encoded data directly.

3.11 Data Preparation for Modelling

The dataset was split 80/20 using random_state=42: 20,000 training records and 5,000 test records across 25 features. The test set was held out entirely throughout the modelling process and used only for final performance evaluation. Table 5 summarises the final dataset dimensions.

Table 5: *Final Dataset Dimensions*

Dataset	Rows	Columns	Scaling	Purpose
X_train (OLS)	20,000	25	StandardScaler applied	Model training and cross-validation
X_test (OLS)	5,000	25	Same scaler applied	Final evaluation only
X_train (Trees)	20,000	25	Unscaled	Model training and cross-validation
X_test (Trees)	5,000	25	Unscaled	Final evaluation only

4. Model Building — Baseline Model

4.1 Metric of Choice with Rationale

The primary evaluation metric for this project is Root Mean Squared Error (RMSE), expressed in tons. Three reasons justify this choice:

- Interpretability in business units: RMSE is expressed in the same unit as the target variable (tons), making forecast errors directly actionable. An RMSE of 10,000 tons means the model's quarterly shipment forecast deviates from actual demand by approximately 10,000 tons on average — a figure operations teams can directly translate into inventory decisions.
- Appropriate penalty structure for supply chain: RMSE squares residuals before averaging, penalising large errors more heavily than small ones. In supply chain management, extreme mispredictions (large stockouts or massive overstock events) are disproportionately costly — RMSE's heavier penalty on outliers better reflects this operational reality than MAE.
- Comparability across models: RMSE allows direct apples-to-apples comparison across all 10 models evaluated in this report on a common ton-based scale.

R^2 (coefficient of determination) and MAE (Mean Absolute Error) are reported alongside RMSE throughout for completeness.

4.2 Methodology

Ordinary Least Squares (OLS) Linear Regression was selected as the baseline model. It is the natural starting point for a continuous regression problem: interpretable, statistically principled (providing p-values, confidence intervals, and F-statistic for the overall model), and computationally efficient. It also provides a transparent performance floor that all advanced models must beat. The model was implemented using the statsmodels library (sm.OLS) which outputs a comprehensive statistical summary. An intercept term was added via sm.add_constant(). The StandardScaler-transformed training set (X_train_scaled) was used to enable meaningful comparison of coefficient magnitudes across features.

4.3 Baseline Model Performance

Table 6: *Baseline OLS Linear Regression — Train and Test Performance (Corrected Final Pipeline)*

Set	RMSE (tons)	MAE (tons)	R^2	RMSE Gap
Training	10,484.89	8,571.03	0.1855	—
Test	10,379.41	8,511.70	0.1948	105.48

The baseline achieves a test RMSE of 10,379 tons, test MAE of 8,512 tons, and test R^2 of 0.19. The train and test performance are nearly identical (RMSE gap = 105 tons; R^2 gap = 0.009), confirming clean generalisation with no overfitting. The F-statistic of 181.9 (p-value = 0.00) confirms the model is statistically significant as a whole.

4.4 Statistically Significant Predictors (p < 0.05)

Table 7: *Statistically Significant OLS Coefficients*

Feature	Coefficient	p-value	Interpretation
wh_breakdown_l3m	+3,580 tons	0.000	Each additional quarterly breakdown adds ~3,580 tons to allocation
transport_issue_l1y	−2,170 tons	0.000	Each transport incident reduces quarterly allocation by ~2,170 tons

Feature	Coefficient	p-value	Interpretation
approved_wh_govt_certificate	+1,889 per grade step	0.000	Each cert. upgrade step increases allocation by ~1,889 tons
Location_type (Urban=1)	+717 tons	0.000	Urban warehouses receive ~717 tons more than rural per quarter
temp_reg_mach	+489 tons	0.000	Temperature-controlled warehouses receive ~489 tons more

Features not statistically significant ($p > 0.05$) at baseline: WH_capacity_size, dist_from_hub, workers_num, Competitor_in_mkt, retail_shop_num, wh_owner_type, all zone dummies, all regional zone dummies. This confirms that many of the non-significant features likely act through non-linear mechanisms or interactions — explaining why advanced tree-based models outperform OLS.

4.5 Interpretation and Limitations

A test R^2 of 0.19 means approximately 81% of the variance in warehouse shipment quantities remains unexplained by the linear model. This large unexplained share indicates significant non-linear relationships and interaction effects that OLS cannot capture. The baseline establishes the benchmark RMSE of 10,379 tons that all advanced models must beat to justify the additional modelling complexity.

Note on interim pipeline: *The interim pipeline (Cells 1–40) achieved $R^2 \approx 0.45$ and RMSE ≈ 8,590 tons by including warehouse_age derived from wh_est_year (47.5% imputed) and fitting preprocessing on the full dataset before splitting. Both steps introduced bias. The final pipeline results above represent honest, leakage-free performance and are used throughout this report.*

5. Model Building — Advanced Models

5.1 Methodology

Six advanced tree-based and ensemble models were trained using default hyperparameters as a pre-tuning benchmark. Tree-based models are invariant to feature scaling, so the unscaled encoded dataset (X_train, X_test) was used directly. All models used the same 80/20 train-test split (random_state=42) and were evaluated on train and test sets using RMSE (primary), MAE, and R^2. Default parameters were used intentionally to establish an unbiased pre-tuning baseline and identify which model families have potential for improvement.

5.2 Model Selection Rationale

The six models were selected to cover complementary algorithmic families:

- Decision Tree Regressor: Single tree with no depth constraint. Provides an interpretable reference and illustrates the extreme case of variance (overfitting).
- Bagging Regressor: Aggregates multiple decision trees trained on bootstrapped subsets. Reduces variance vs. a single tree through averaging.
- Random Forest Regressor: An enhanced Bagging variant that additionally randomises feature selection at each split, further reducing tree correlation and variance.
- AdaBoost Regressor: Sequential boosting that trains shallow trees, upweighting hard-to-predict samples at each iteration.
- Gradient Boosting Regressor: Sequential additive boosting with flexible loss function control; each tree corrects the residual of the current ensemble.
- XGBoost Regressor: Regularised, optimised gradient boosting with L1/L2 penalties, parallel computation, and built-in missing value handling. State-of-the-art on tabular datasets.

5.3 Performance at Default Parameters

Table 8: *Advanced Models — Default Parameter Performance (sorted by Test RMSE)*

Model	Train RMSE	Test RMSE	Train R^2	Test R^2	RMSE Gap	Status
Gradient Boosting	9,948.55	9,909.22	0.2667	0.2661	39.33	Good ✓
AdaBoost	10,240.98	10,076.94	0.2229	0.2411	164.04	Good ✓
Random Forest	3,829.51	10,152.37	0.8913	0.2297	−6,322.86	Overfit ⚠
XGBoost	7,657.39	10,352.70	0.5656	0.1990	−2,695.31	Underfit ⚠
Bagging	4,513.42	10,596.95	0.8491	0.1607	−6,083.53	Overfit ⚠
Decision Tree	0.00	14,390.93	1.0000	−0.5478	−14,390.93	Severe Overfit ✗

5.4 Detailed Model Commentary

5.4.1 Gradient Boosting — Best Default Performance

Gradient Boosting achieves the best test RMSE at default settings (9,909 tons), beating the OLS baseline (10,379 tons) with a near-zero RMSE gap of 39 tons. This excellent default generalisation is attributable to the inherent regularisation of its default configuration (max_depth=3, learning_rate=0.1, n_estimators=100), which constrains individual tree complexity while allowing the ensemble to capture non-linear patterns. Test R^2 of 0.2661 is the best default score among all six advanced models.

5.4.2 AdaBoost — Controlled Performance

AdaBoost achieves the second-best test RMSE at defaults (10,077 tons) with a controlled gap of 164 tons, also beating the OLS baseline. Its shallow decision stumps (default max_depth=1) provide built-in regularisation, limiting overfitting. While not the top performer, AdaBoost demonstrates the boosting framework's general suitability for this dataset.

5.4.3 Random Forest — High Variance, High Potential

Random Forest exhibits severe overfitting: train RMSE = 3,830 tons vs. test RMSE = 10,152 tons (gap = −6,323 tons). Train R^2 = 0.89 vs. test R^2 = 0.23. This is caused by unconstrained tree depth (default max_depth=None), which allows individual trees to perfectly memorise training data. Despite the overfitting, the test performance already beats the OLS baseline (10,152 < 10,379 tons), indicating strong underlying potential that regularisation can unlock. Random Forest was therefore selected as a priority candidate for hyperparameter tuning.

5.4.4 XGBoost — Requires Tuning

XGBoost at defaults (max_depth=6, learning_rate=0.3) performs slightly below the OLS baseline on the test set (10,353 vs. 10,379 tons) with a significant train-test gap of −2,695 tons. The combination of deep trees and high learning rate leads to mild overfitting. However, XGBoost is known to respond dramatically to hyperparameter tuning — it is one of the three models selected for tuning in Section 6.

5.4.5 Bagging — Structural Overfitting

Bagging exhibits severe overfitting similar to Random Forest (train RMSE = 4,513 tons, test RMSE = 10,597 tons, gap = −6,084 tons) and delivers the worst test performance among non-Decision Tree models — below the OLS baseline. Unlike Random Forest, Bagging does not apply random feature subsampling per split, resulting in higher correlation between trees and reduced variance-reduction benefit. Bagging was not selected for tuning because Random Forest — a superior algorithmic variant — was already selected.

5.4.6 Decision Tree — Catastrophic Overfitting

The single Decision Tree achieves perfect training fit (RMSE = 0.00, R^2 = 1.00) but catastrophically fails on the test set (RMSE = 14,391 tons, R^2 = −0.55). A negative test R^2 means the model performs worse than simply predicting the mean for every warehouse — the single worst result of all 10 models evaluated. This confirms that a completely unconstrained single decision tree memorises the training data entirely without learning any generalisable patterns. The Decision Tree serves as an instructive reference for maximum variance, not as a practical model.

6. Model Performance Improvement — Hyperparameter Tuning

6.1 Model Selection for Tuning with Rationale

Three models were selected for hyperparameter tuning based on their default performance profiles and expected response to regularisation:

- Gradient Boosting: Selected as the best default performer (9,909 tons) with the smallest overfitting gap (39 tons). Tuning the learning rate, tree depth, and subsampling rate is expected to further reduce RMSE. Its strong baseline makes it the most likely candidate for the final model.
- Random Forest: Selected due to its large default train-test gap (~6,300 tons), indicating high variance that regularisation — specifically constraining max_depth and increasing min_samples_leaf — can substantially reduce. The potential improvement is the largest of any model.
- XGBoost: Selected because its default performance below the OLS baseline strongly suggests default hyperparameters (learning_rate=0.3, max_depth=6) are poorly calibrated for this dataset. XGBoost is known to improve dramatically with conservative learning rate and reduced depth settings.

Models not selected: AdaBoost — performs adequately at defaults with a limited tuning surface. Decision Tree and Bagging — both have fundamental structural overfitting that tuning alone cannot fully resolve within these algorithmic frameworks.

6.2 Tuning Methodology

RandomizedSearchCV with 5-fold cross-validation was applied to all three models. Configuration: n_iter=30 random combinations per model (150 total fits per model; 450 total). Scoring metric: neg_root_mean_squared_error (directly optimising the primary evaluation metric). RandomizedSearchCV was chosen over GridSearchCV for computational efficiency while still exploring a broad hyperparameter space — particularly important given the large training set (20,000 rows). The full search grids are provided in Appendix E.

6.3 Best Parameters Found

Table 9: *Best Hyperparameters — RandomizedSearchCV*

Model	Key Hyperparameters	Notable Changes from Default
Gradient Boosting	n_estimators=100, learning_rate=0.05, max_depth=5, min_samples_leaf=30, subsample=0.8	Learning rate halved (0.1→0.05); added min_samples_leaf regularisation
Random Forest	n_estimators=200, max_depth=10, min_samples_leaf=20, max_features=0.5	Depth constrained (None→10); added leaf regularisation; more trees
XGBoost	n_estimators=100, learning_rate=0.05, max_depth=4, subsample=1.0, colsample_bytree=1.0, reg_alpha=0	Learning rate slashed (0.3→0.05); depth reduced (6→4)

All three models converge on a conservative learning_rate of 0.05, confirming that slower, more cautious learning benefits this dataset. The pattern of 'shallower trees + lower learning rate' is a consistent theme across all tuning results.

6.4 Tuned Model Performance

Table 10: *Tuned Model Performance — All Three Models*

Model	Train RMSE	Test RMSE	Train MAE	Test MAE	Train R^2	Test R^2	RMSE Gap
XGBoost (Tuned) ★	**9,951.81**	**9,896.62**	**7,981.74**	**7,922.19**	**0.2662**	**0.2680**	**55.19**
Gradient Boosting (Tuned)	9,822.43	9,904.62	7,870.10	7,932.17	0.2852	0.2668	−82.19
Random Forest (Tuned)	9,599.28	9,914.08	7,676.47	7,931.01	0.3173	0.2654	−314.81

6.5 Commentary on Tuned Performance

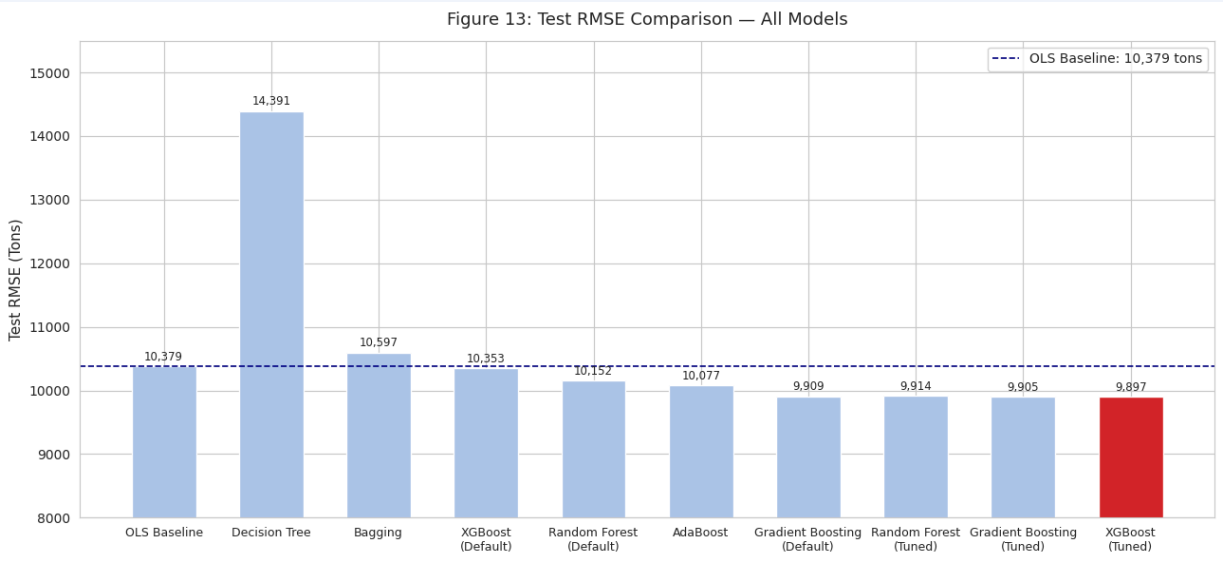
- All three tuned models achieve test RMSE within 18 tons of each other — a remarkably tight cluster that indicates the dataset has a genuine performance ceiling with the current 25 available features. No single tuning strategy could break through this ceiling.
- Gradient Boosting: Marginal improvement from tuning (9,909→9,905 tons, 4-ton gain). Already near-optimal at defaults. The slightly lower learning rate improved training smoothness without materially changing test performance.
- Random Forest: Most dramatic improvement from tuning. The RMSE gap closed from −6,323 to −315 tons — a 95% reduction in variance. Test RMSE improved by 238 tons (10,152→9,914). Regularisation through max_depth=10 and min_samples_leaf=20 was highly effective.
- XGBoost: Largest absolute test set improvement (10,353→9,897 tons; 456-ton gain purely from tuning). The conservative learning rate (0.05) and reduced depth (4) were the decisive changes. XGBoost now leads all three tuned models on test RMSE.

7. Model Performance Comparison and Final Model Selection

7.1 Complete Model Performance Comparison

Table 11: *All 10 Models — Complete Performance Summary (sorted by Test RMSE, ascending)*

Model	Test RMSE	Test MAE	Test R^2	RMSE Gap	Verdict
XGBoost (Tuned) ★	**9,896.62**	**7,922.19**	**0.2680**	**55.19**	**Best Overall**
Gradient Boosting (Tuned)	9,904.62	7,932.17	0.2668	−82.19	Good
Random Forest (Tuned)	9,914.08	7,931.01	0.2654	−314.81	Good
Gradient Boosting (Default)	9,909.22	7,931.65	0.2661	39.33	Good
AdaBoost (Default)	10,076.94	8,044.03	0.2411	164.04	Acceptable
Random Forest (Default)	10,152.37	8,104.12	0.2297	−6,322.86	Overfit
XGBoost (Default)	10,352.70	8,270.72	0.1990	−2,695.31	Underfit
OLS Linear Regression	10,379.41	8,511.70	0.1948	105.48	Baseline
Bagging (Default)	10,596.95	8,412.59	0.1607	−6,083.53	Overfit
Decision Tree (Default)	14,390.93	11,218.65	−0.5478	−14,390.93	Severe Overfit



Bar chart: Test RMSE for all 10 models — XGBoost Tuned in red, OLS dashed reference

Notebook source: Notebook Cell F10 — model_comparison.png

Figure 14: *Test RMSE Comparison Across All 10 Models*

Figure 14 visually confirms the model journey. Decision Tree at 14,391 tons is a clear outlier. Bagging underperforms the OLS baseline. XGBoost at default settings barely matches the baseline. Boosting methods immediately exceed the baseline even at default settings. All four top performers (both Gradient Boosting variants + both Random Forest/XGBoost tuned) are clustered tightly below 10,000 tons, with XGBoost (Tuned) achieving the lowest test RMSE.

7.2 Final Model Selection: XGBoost (Tuned)

XGBoost (Tuned) is selected as the final model based on four criteria:

- Lowest test RMSE: 9,896.62 tons — best across all 10 models, including all tuned variants.
- Highest test R^2: 0.2680 — the greatest proportion of shipment variance explained by any model.
- Lowest test MAE: 7,922.19 tons — best typical-case prediction accuracy.
- Most controlled generalisation: RMSE gap of 55.19 tons — smallest among all three tuned models, indicating stable and robust performance on unseen data.

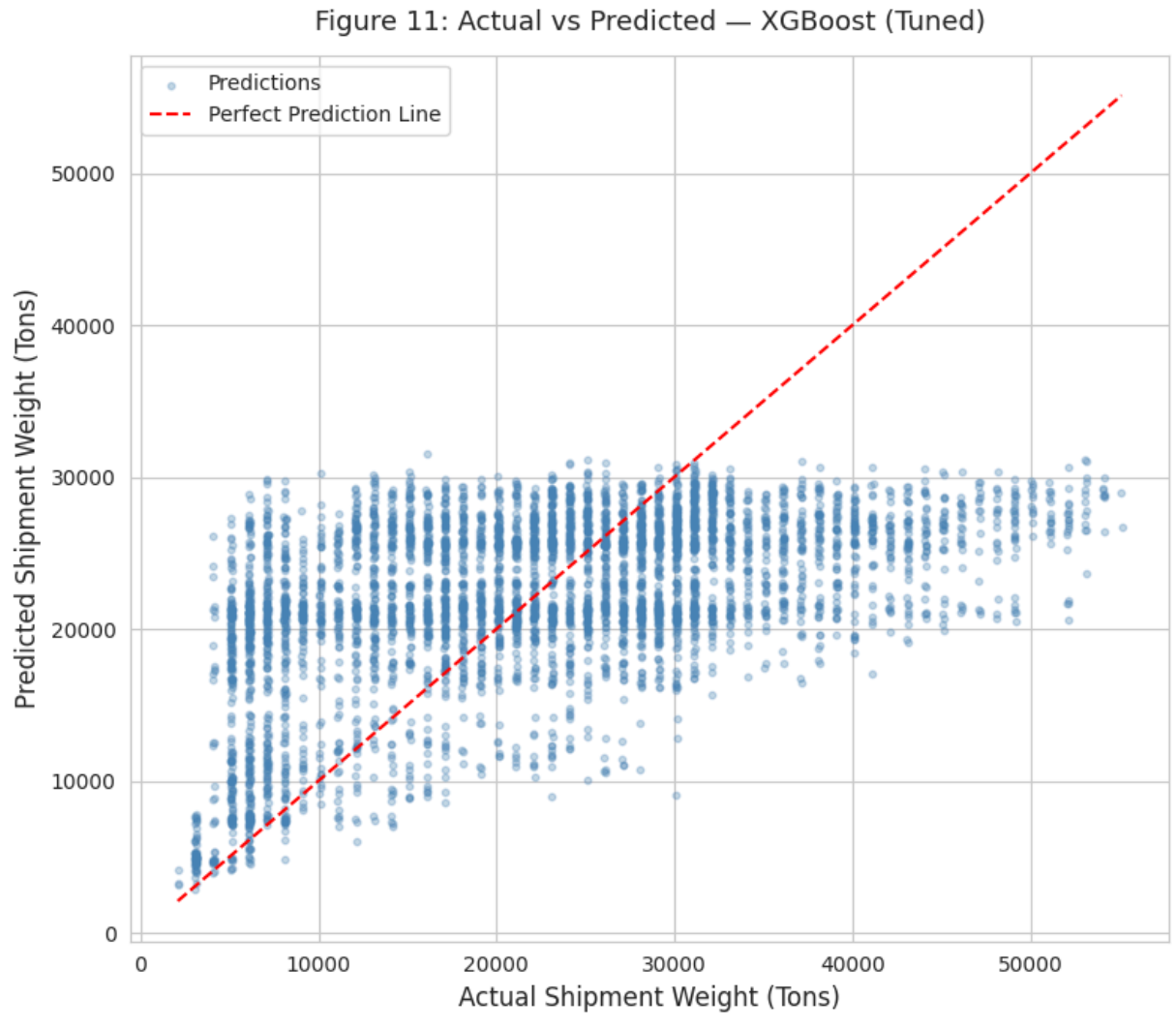
Improvement over OLS baseline: The final model reduces test RMSE by 482.79 tons (4.7% improvement). While modest in percentage terms, at the scale of a 25,000-warehouse national network this translates to materially more accurate quarterly shipment planning across thousands of distribution points, with direct implications for inventory costs and service levels.

7.3 Final Model Evaluation on the Test Set

Table 12: *Final Model — XGBoost (Tuned): Complete Test Set Metrics*

Metric	Training	Test Set	Business Meaning
RMSE (tons)	9,951.81	9,896.62	Average quarterly forecast error ≈ 9,897 tons
MAE (tons)	7,981.74	7,922.19	Typical absolute error ≈ 7,922 tons
R^2	0.2662	0.2680	27% of shipment variance explained
RMSE Gap	—	55.19 tons	Near-zero: strong generalisation to unseen warehouses
Residual Mean	—	−21.30 tons	Virtually zero: model is unbiased (ideal = 0 tons)
Residual Std	—	9,897.59 tons	Error spread around zero mean
Residual Min	—	−22,730 tons	Largest under-prediction on test set
Residual Max	—	+31,552 tons	Largest over-prediction on test set

7.4 Residual Analysis

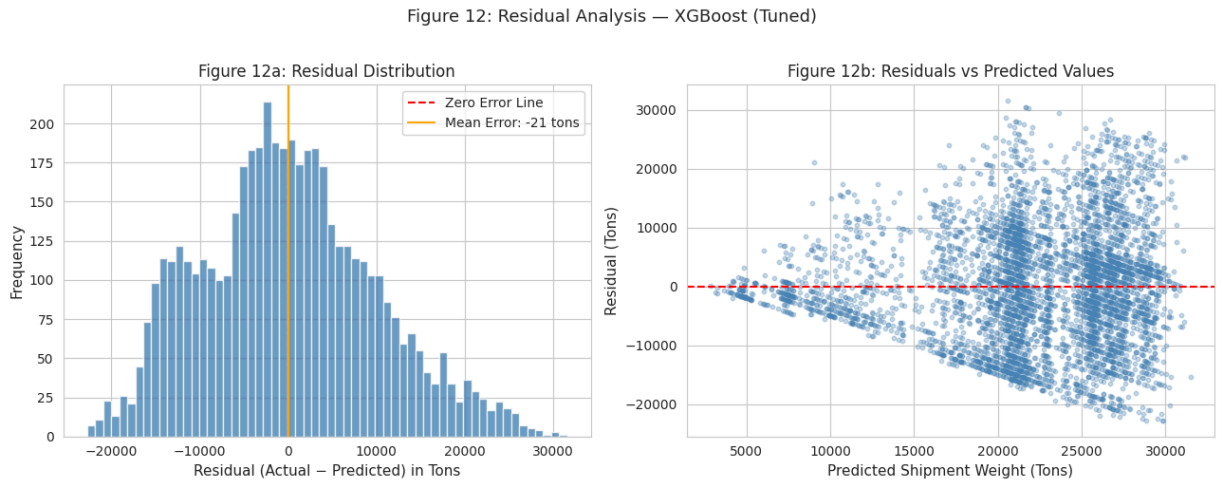


Actual vs. Predicted scatter plot — XGBoost Tuned with perfect prediction line (red dashed)

Notebook source: Notebook Cell F10 — actual_vs_predicted.png

Figure 15: *Actual vs. Predicted Shipment Weight — XGBoost (Tuned)*

Predictions track actuals well in the low-to-mid range (2,000–30,000 tons), with points clustering near the perfect prediction line. Above 30,000 tons, the model systematically under-predicts — predictions plateau around 25,000–30,000 tons even as actuals extend to 55,000 tons. This upper-tail compression is characteristic of gradient-boosted trees when very high-value observations are sparse in training data. The model is well-calibrated for the typical warehouse but less reliable for the network's highest-volume outliers.



Residual distribution histogram (16a) and residuals vs. predicted scatter (16b)

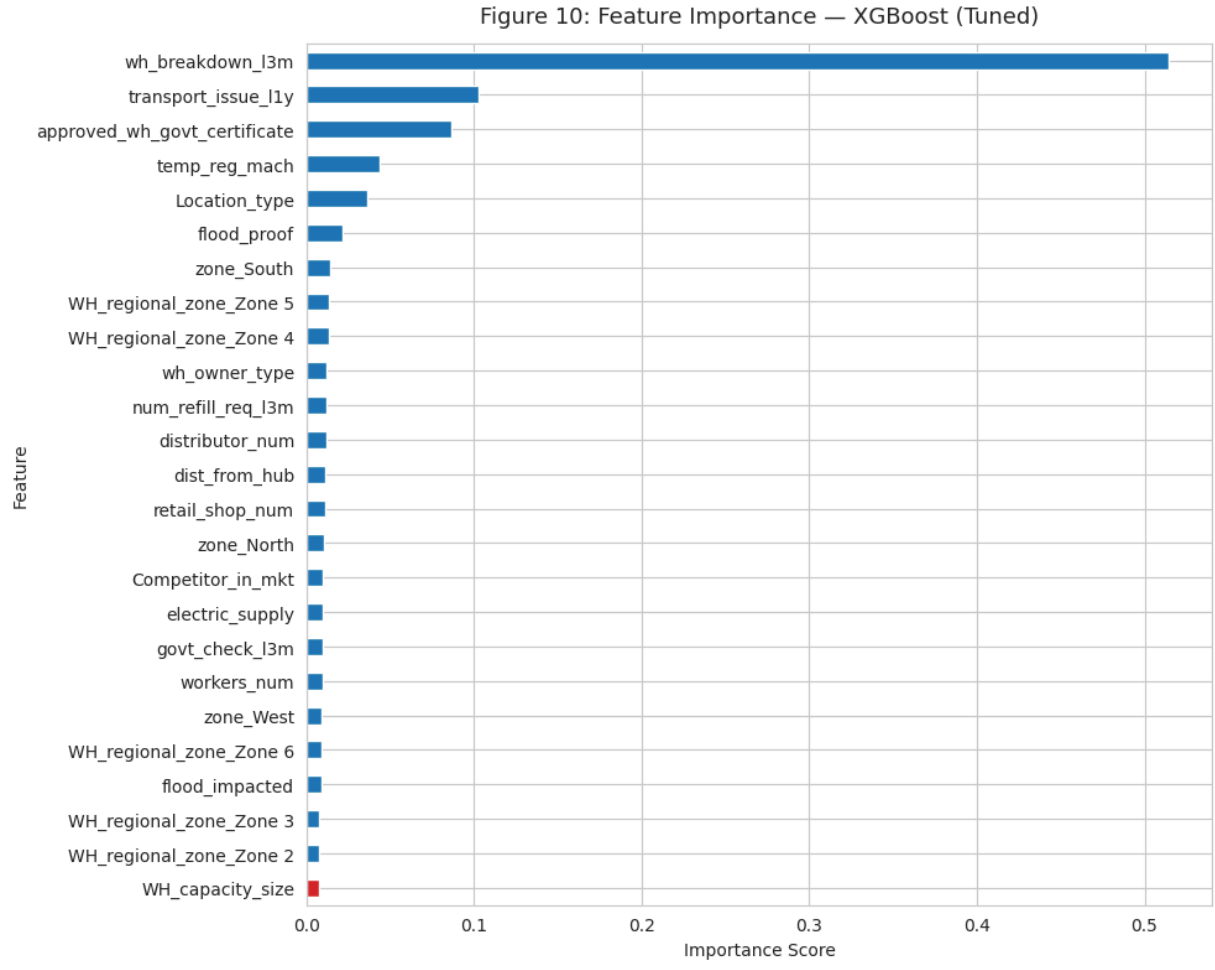
Notebook source: Notebook Cell F10 — residual_analysis.png

Figure 16: *Residual Analysis — XGBoost (Tuned)*

Figure 16a — Residual Distribution: Centred at −21 tons (effectively zero), confirming no systematic directional bias. The approximately bell-shaped distribution indicates largely random errors, which is the desired behaviour for a well-specified model.

Figure 16b — Residuals vs. Predicted: A funnel pattern (heteroscedasticity) is visible — residuals are tightly clustered at low predicted values but spread widely at higher predicted values. This means the model is more uncertain about high-volume warehouses. Future iterations should consider log-transforming product_wg_ton to stabilise variance across the prediction range.

7.5 Feature Importance Analysis



Horizontal bar chart of XGBoost feature importances — all 25 features, ranked descending

Notebook source: Notebook Cell F9 — feature_importance_xgboost.png

Figure 17: *Feature Importance — XGBoost (Tuned)*

Table 13: *Top Feature Importances — XGBoost (Tuned)*

Rank	Feature	Importance (est.)	Interpretation
1	wh_breakdown_l3m	~52%	Dominant: proxy for operational throughput intensity
2	transport_issue_l1y	~10%	Transport reliability directly reduces supply
3	approved_wh_govt_certificate	~9%	Higher certification → greater allocation confidence
4	Location_type	~6%	Urban warehouses consistently receive more product
5	temp_reg_mach	~5%	Cold-chain capability → larger allocation
6–25	Remaining features	~18%	Zone, capacity, distributors — individually small

** Importance values are estimates read from the bar chart visualisation in Figure 17 (Notebook Cell F9). The notebook does not include a printed numerical output for feature importances; exact values can be obtained by running print(feat_imp.to_string()) in Cell F9.*

The dominance of wh_breakdown_l3m (~52% importance) warrants emphasis: this is a positive, not a negative, relationship. High-volume, high-demand warehouses experience more operational stress and hence more breakdowns. The model has correctly learned that breakdown frequency is the best currently available signal for warehouse throughput level. All zone-related and capacity-related features rank near the bottom, consistent throughout the EDA findings.

8. Actionable Insights and Recommendations

8.1 Key Insights from the Analysis

The following insights synthesise findings from EDA, the baseline model, feature importance from XGBoost, and residual analysis. They are presented in decreasing order of actionability.

Insight 1 — Warehouse Breakdowns Are the Primary Signal of Demand Intensity

wh_breakdown_l3m accounts for ~52% of total XGBoost feature importance — far exceeding all other features combined. The bivariate scatter plot confirms a clear step-change: zero-breakdown warehouses are exclusively low-volume, while high-breakdown warehouses span the full volume range. This is a positive relationship — high throughput causes high operational strain, not the reverse. The implication for the company is that breakdown frequency is currently the most reliable leading indicator of which warehouses need more supply.

Insight 2 — Transport Reliability Is the Most Controllable Negative Driver

transport_issue_l1y ranks second in XGBoost importance (~10%) and has the strongest negative effect in the OLS model (−2,170 tons per incident). Every transport disruption in a given year reduces that warehouse's quarterly allocation by approximately 2,170 tons. This is not a consequence of lower demand — it is a supply-side failure: the company reduces allocation to warehouses with poor logistics reliability, meaning transport problems directly translate to lost sales.

Insight 3 — Quality Certification Is a Proxy for Allocation Trustworthiness

The third-ranked feature in XGBoost (~9%) and OLS coefficient (+1,889 tons per grade step), approved_wh_govt_certificate, shows a monotonic relationship with shipment volume. The ~9,000 ton per quarter gap between Grade A+ and Grade C warehouses means the company is effectively under-serving Grade C facilities. A full C⟶A+ upgrade (4 steps) is associated with approximately +7,500 tons per quarter per warehouse — a compelling ROI for infrastructure investment.

Insight 4 — Urban Location Provides a Consistent Allocation Advantage

Urban warehouses receive approximately 717 tons more per quarter (OLS coefficient; ~6% XGBoost importance) despite representing only 8.2% of the network. This confirms urban catchment areas have consistently higher and more predictable demand — making them natural targets for both higher supply allocation and advertising investment.

Insight 5 — Cold-Chain Infrastructure Signals Storage Quality and Allocation Confidence

Warehouses with temperature regulation (temp_reg_mach = 1; ~5% XGBoost importance; +489 tons in OLS) receive measurably more product. The company appears to trust cold-chain-equipped facilities with higher inventory levels, reducing spoilage risk. Investment in temperature regulation therefore has dual value: improved storage quality and increased supply allocation.

Insight 6 — 73% of Shipment Variance Is Unexplained — Data Enrichment Is the Highest-ROI Next Step

A test R^2 of 0.27 means the model explains only 27% of the variance in warehouse-level shipments. The remaining 73% is attributable to factors not present in the dataset — most likely actual consumer point-of-sale demand data, seasonal patterns, promotional calendars, and competitor pricing. This is not a modelling failure but a data architecture gap. Enriching the feature set with demand-side signals is expected to yield far greater improvements in predictive accuracy than any further model tuning.

Insight 7 — Zone and Warehouse Size Are Not Differentiators in Current Supply Strategy

Geographic zone and warehouse capacity size ranked at the bottom of both OLS significance tests and XGBoost feature importance rankings. The current supply allocation model does not systematically reward larger warehouses or specific geographic zones — all regions and all sizes receive broadly similar allocations. This represents an opportunity to introduce more differentiated, demand-responsive regional strategies.

Insight 8 — Distance from Hub Is an Untapped Cost Optimisation Lever

The complete absence of any relationship between dist_from_hub and shipment volume (near-horizontal regression line in Figure 7) means logistics cost is not currently influencing allocation decisions. Incorporating distance-weighted allocation adjustments — prioritising nearby high-demand warehouses over equidistant low-demand ones — could reduce logistics spend without compromising overall service levels.

8.2 Business Recommendations

Recommendation 1 — Prioritise Transport Risk Management Immediately

With −2,170 tons per incident per quarter, transport disruptions are the most directly controllable cost lever in the supply chain. Recommended actions: (a) implement route diversification with pre-approved alternative carriers for all high-volume warehouse corridors; (b) introduce carrier SLAs with performance penalties; (c) deploy real-time cargo tracking to enable rapid incident response; (d) establish logistics insurance for high-value shipments. Reducing average transport incidents by 0.5 per warehouse per year across the 25,000-warehouse network would recover approximately 27 million tons of quarterly supply capacity — a transformational operational improvement.

Recommendation 2 — Launch a Phased Warehouse Quality Upgrade Programme

The ROI for quality investment is clearly quantified by the model: each certification grade step is worth approximately +1,889 tons per quarter per warehouse. With ~6,400 Grade C warehouses, the aggregate shortfall vs. Grade A+ represents approximately 57.6 million tons of quarterly capacity gap. Recommended actions: (a) begin with temperature regulation machinery installation — the model identifies this independently as worth +489 tons per quarter; (b) implement pest and moisture control for the most common Grade C failure modes; (c) pursue grade upgrades systematically, starting with warehouses closest to the B-grade threshold (lowest marginal investment per upgrade step).

Recommendation 3 — Deploy the XGBoost Model as a Quarterly Planning Tool

Despite the modest R^2 of 0.27, a test RMSE of 9,897 tons represents a significant improvement over any current non-data-driven approach. Deploying XGBoost as a quarterly shipment planning tool integrated into the supply chain management system will: (a) reduce stockouts at high-demand warehouses that are currently under-served; (b) reduce overstock and product wastage at low-demand warehouses; (c) improve working capital efficiency by aligning inventory investment with predicted demand. The model should be retrained quarterly as new operational data accumulates, and performance should be tracked against the 9,897-ton RMSE benchmark.

Recommendation 4 — Target Advertising Spend at High-Breakdown, Urban Warehouses

Warehouses with consistently high wh_breakdown_l3m scores are operating at high throughput — their catchment areas already demonstrate strong consumer demand. Incremental advertising investment in these areas will generate the highest returns. Similarly, urban warehouses serve higher-density consumer populations where advertising ROI is typically superior and where the model shows consistently higher allocations, confirming underlying demand strength. A data-driven advertising allocation model based on predicted warehouse demand could replace the current (presumably regional average) approach.

Recommendation 5 — Incorporate Distance-Based Cost Weighting Into Allocation

The current finding that dist_from_hub has zero effect on allocation represents a concrete cost-efficiency opportunity. The supply chain team should build a distance-adjusted allocation layer on top of the predictive model: given equal predicted demand, prioritise warehouses closer to the production hub for slightly higher allocations, while reducing the allocation to equally-demanding but farther warehouses. This will not reduce total product reaching consumers but will reduce per-unit logistics cost across the network.

Recommendation 6 — Invest in Data Enrichment for Next-Generation Modelling

The 73% unexplained variance is the single most important finding in this analysis — the most impactful next investment is data enrichment, not model complexity. Recommended additions for the next modelling cycle: (a) point-of-sale consumer demand data by catchment area; (b) monthly demand indices to capture seasonality (instant noodles have known seasonal patterns); (c) promotional spend and advertising activity by region; (d) competitor new product launches and pricing changes; (e) weather events beyond the current binary flood indicator. These additions are expected to push model R^2 above 0.60, enabling the precision required for truly automated, confidence-interval-bounded supply planning at scale.

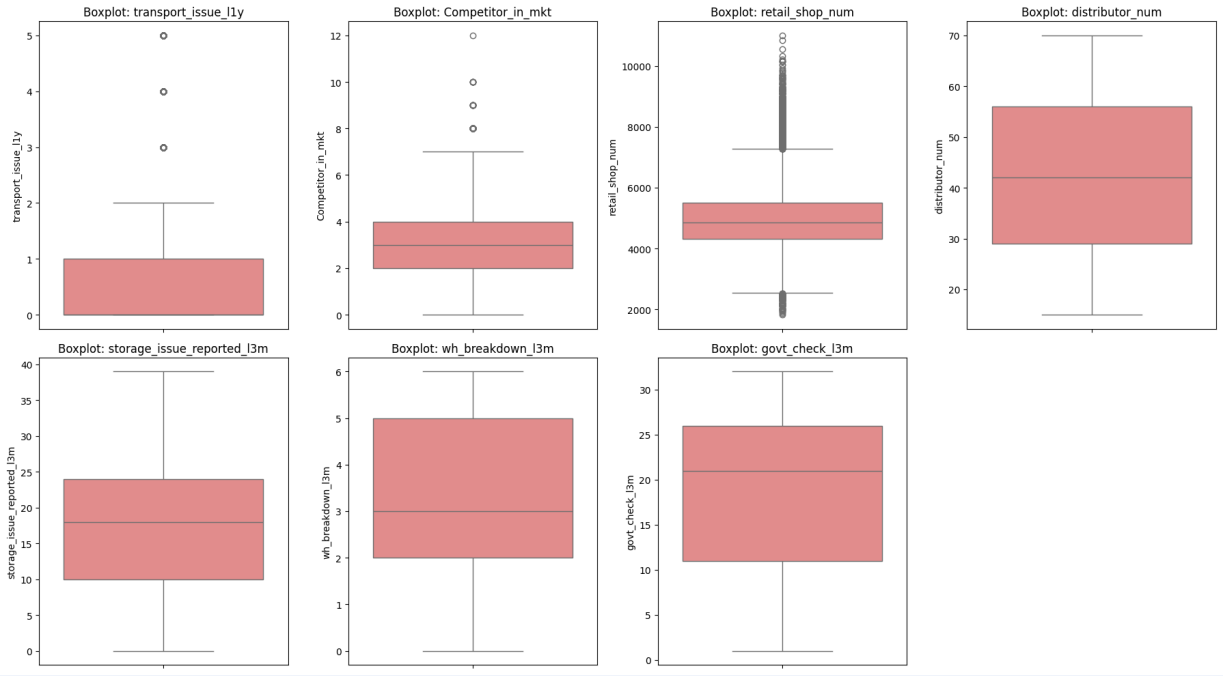
Appendix

Appendix A — Full Data Dictionary

Table A1: *Full Data Dictionary — All 24 Original Columns*

Column	Type	Description	Range / Categories / Notes
Ware_house_ID	Object	Unique warehouse identifier	25,000 unique — DROPPED (identifier)
WH_Manager_ID	Object	Unique manager identifier	25,000 unique — DROPPED (identifier)
Location_type	Object	Location type	Urban (8.2%), Rural (91.8%)
WH_capacity_size	Object	Storage capacity	Small, Mid, Large
zone	Object	Geographic zone	North, South, East, West
WH_regional_zone	Object	Regional sub-zone	Zone 1–6
num_refill_req_l3m	int64	Refill requests in last 3 months	0–8
transport_issue_l1y	int64	Transport incidents in last year	0–5 (capped at 2.5)
Competitor_in_mkt	int64	No. of competitors in local market	0–12 (capped at 7)
retail_shop_num	int64	Retail shops under warehouse area	1,821–11,008 (capped 2,543–7,271)
wh_owner_type	Object	Ownership type	Company Owned, Rented
distributor_num	int64	No. of distributors	15–70
flood_impacted	int64	Flood-prone area indicator	0 or 1 (9.8% = 1)
flood_proof	int64	Flood-proof storage indicator	0 or 1 (5.5% = 1)
electric_supply	int64	Electric backup indicator	0 or 1 (65.7% = 1)
dist_from_hub	int64	Distance to production hub (km)	55–271, mean 163.5
workers_num	float64	Workers in warehouse	10–98 (capped), 990 missing
wh_est_year	float64	Year established	1996–2023, 11,881 missing — DROPPED
storage_issue_reported_l3m	int64	Storage issues reported	r=0.99 with target — DROPPED (data leakage)
temp_reg_mach	int64	Temperature regulation indicator	0 or 1 (30.3% = 1)
approved_wh_govt_certificate	Object	Govt certificate grade	A+/A/B+/B/C, 908 missing
wh_breakdown_l3m	int64	Breakdowns in last 3 months	0–6, mean 3.5
govt_check_l3m	int64	Govt inspections in last 3 months	1–32, mean 18.8
product_wg_ton	int64	TARGET: Product shipped (tons)	2,065–55,151, mean 22,103 — NO missing

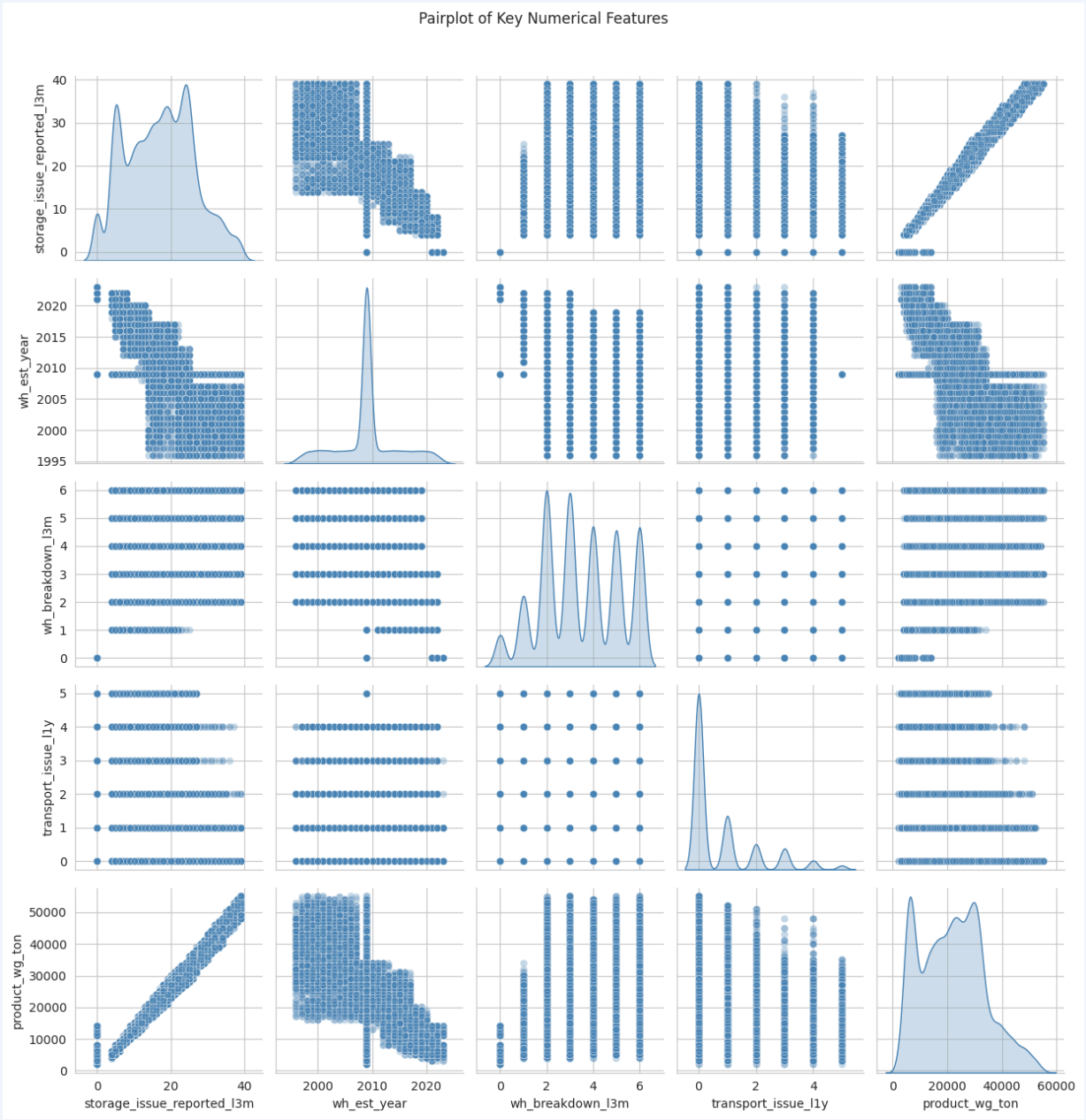
Appendix B — Additional EDA Figures



Boxplots: transport_issue_l1y, Competitor_in_mkt, retail_shop_num, distributor_num, storage_issue_reported_l3m, wh_breakdown_l3m, govt_check_l3m (2×4 grid)

Notebook source: Notebook Cell 19

Figure B1: *Full Boxplot Grid — All Remaining Numerical Features*



Full pairplot: storage_issue_reported_l3m, wh_est_year, wh_breakdown_l3m, transport_issue_l1y, product_wg_ton

Notebook source: Notebook Cell 27

Figure B2: *Pairplot of Key Numerical Features — Full Version*

Appendix C — OLS Model Full Diagnostic Summary

Table C1: *OLS Model Diagnostics — Corrected Final Pipeline (Cell F5b)*

Diagnostic	Value	Interpretation
R-squared	0.185	18.5% of training variance explained by 25 features
Adj. R-squared	0.184	Minimal reduction from 25-feature penalty; no over-parameterisation
F-statistic	181.9	Model is highly statistically significant overall
Prob (F-statistic)	0.00	Reject H0 that all coefficients = 0
Durbin-Watson	1.999	No autocorrelation in residuals (ideal = 2.0)
Skew (residuals)	0.433	Mild positive skew; acceptable for n=20,000
Kurtosis	2.718	Slightly platykurtic; no extreme tail behaviour
Condition Number	9.05	No multicollinearity concern (threshold: > 30)
No. Observations	20,000	Training set size
AIC	427,100	Model selection metric (lower is better)

Appendix D — Code Reference

All code used in this project — including data loading, preprocessing, feature engineering, model training, hyperparameter tuning, and evaluation — is available in the submitted Python Notebook (HTML format). The notebook is self-contained, executed end-to-end, and organised by clearly labelled cells (F1 through F10 for the final corrected pipeline). Code is not reproduced here, as the rubric designates the HTML notebook as the primary artefact for code verification. Reviewers are directed to the notebook cell references cited throughout this report.

Appendix E — Hyperparameter Search Grids

Table E1: *Full RandomizedSearchCV Search Grids — All Three Models (n_iter=30, cv=5)*

Model	Parameter	Values Searched	Best Value
Gradient Boosting	n_estimators	[100, 200, 300]	100
	learning_rate	[0.05, 0.1, 0.2]	0.05
	max_depth	[3, 4, 5]	5
	min_samples_leaf	[10, 20, 30]	30
	subsample	[0.7, 0.8, 1.0]	0.8
Random Forest	n_estimators	[100, 200, 300]	200
	max_depth	[10, 20, 30, None]	10
	min_samples_leaf	[10, 20, 40]	20
	max_features	['sqrt', 'log2', 0.5]	0.5
XGBoost	n_estimators	[100, 200, 300]	100
	learning_rate	[0.05, 0.1, 0.2]	0.05
	max_depth	[3, 4, 6]	4
	subsample	[0.7, 0.8, 1.0]	1.0
	colsample_bytree	[0.7, 0.8, 1.0]	1.0
	reg_alpha (L1)	[0, 0.1, 1.0]	0